Implementation of Shrinkage Estimators for Cosmological Precision Matrices

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Abstract. The Ensemble Kalman Filter (EnKF) and cosmological parameter inference share a common statistical bottleneck: both require the inverse of a covariance matrix estimated from a small number of realizations. We embed the direct precision shrinkage framework of Bodnar et al. inside the dual formulation of the EnKF, recomputing the shrinkage coefficients at every assimilation cycle, and benchmark six estimators (four direct precision shrinkage targets, Ledoit–Wolf covariance shrinkage, and NERCOME) on two systems with opposite precision structure: Lorenz 96 with dense off-diagonal precision and BOSS DR12 power-spectrum mocks with near-diagonal precision. Two findings emerge. First, target-truth alignment is decisive: the direct precision methods achieve the lowest Frobenius loss on cosmological data but fail catastrophically on Lorenz 96, where unclamped β^* values reach 10^8–10^9 and clamping occurs in over 80% of cycles. Second, standalone estimation accuracy does not imply good filtering: Ledoit–Wolf gives the worst standalone precision estimates on cosmological data yet achieves the lowest EnKF RMSE, because its low cycle-to-cycle variance compounds favourably over hundreds of assimilation steps. Variance reduction, not bias reduction, is the dominant criterion for sequential filtering. We also propose a domain-agnostic scaled-identity target that sits between the plain identity and the domain-specific cosmological target.

Keywords: Data Assimilation · Ensemble Kalman Filter · Shrinkage Estimation · Precision Matrix · Cosmological Covariance

1 Introduction

Ensemble Kalman filters require the inverse of the background error covariance matrix, but in most operational settings the ensemble is far too small to estimate that matrix reliably. When the state dimension n exceeds the ensemble size N_e, the sample covariance is rank-deficient and cannot be inverted at all [11]; even when $N_e > n$, sampling noise introduces spurious correlations that contaminate the precision matrix [1]. The same problem appears in cosmological data analysis, where the number of mock catalogs available to estimate power spectrum covariances is often comparable to the data dimension [8].

Shrinkage estimation offers a practical remedy: combine the noisy sample estimate with a structured, low-variance target matrix, accepting a small bias in exchange for substantial variance reduction. Most of the established literature operates in covariance space. Ledoit and Wolf [14] derived a closed-form optimal intensity for shrinking toward a scaled identity; Pope and Szapudi [21] introduced shrinkage to cosmology; Chen et al. [3] refined it with an oracle-approximating estimator. Bodnar et al. [2] took a different route, shrinking the precision matrix directly rather than first shrinking the covariance and then inverting. Looijmans et al. [15] applied this direct precision framework to BOSS DR12 mock catalogs, showing that a domain-specific (cosmological) target outperforms generic alternatives in a standalone setting.

None of the existing approaches is without practical drawbacks. Covariance-space shrinkage minimizes $\mathbb{E}\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_F^2$, not the precision-space loss that governs the EnKF update; the inverse of a covariance-optimal estimator amplifies errors in small eigenvalues, which are exactly the directions where regularization is needed. Target-free methods such as NERCOME [13] buy robustness at the cost of repeated bootstrap eigendecompositions that are prohibitive inside a sequential filter. Sparse-precision approaches (graphical lasso [6], neighborhood selection [17]) require a sparsity assumption that is not satisfied by either Lorenz-like dense precision or nearly-diagonal power-spectrum precision. Direct precision shrinkage [2] removes the inversion bias but is only defined for invertible sample covariances and its published target menu is limited to identity, eigenvalue-based, and domain-specific (cosmological) options. Localization [7] side-steps the problem by imposing sparsity but destroys long-range correlations and invalidates the Wishart assumption underlying the shrinkage formulas.

Two concrete gaps remain. First, Looijmans et al. [15] applied these estimators to standalone precision estimation, not inside a sequential filter. The EnKF recomputes the precision matrix at every assimilation cycle, and errors in the precision estimate feed back through the forecast model into subsequent cycles. It is unclear whether an estimator that performs well in a one-shot setting will also perform well when applied repeatedly across hundreds of cycles. Second, the direct-precision target menu lacks a middle ground between the plain identity (ignores variable-specific scales) and the cosmological target (requires domain knowledge), leaving practitioners without a domain-agnostic but scale-aware option.

Contributions. We address both gaps and make the following contributions. (i) We embed the Bodnar et al. [2] direct precision shrinkage framework inside the dual EnKF formulation, recomputing the shrinkage coefficients at each assimilation cycle from the current ensemble, and integrate six estimators into the TEDA toolbox [19]. (ii) We propose a *scaled identity target* $\boldsymbol{\Pi}_0 = \mathrm{diag}(1/(2\sigma_i^2))$ derived from the empirical per-variable variances of the sample covariance, providing a domain-agnostic alternative to the cosmological target. (iii) We benchmark the six estimators on two systems with opposite precision structure—Lorenz 96 [16] ($n = 40$, dense off-diagonal precision) and BOSS DR12 mocks [15] ($d = 18$, near-diagonal precision)—and show that (a) target-truth alignment determines

direct precision shrinkage performance, with the methods failing on Lorenz 96 and succeeding on cosmological data, and (b) good standalone estimation does not imply good filtering, because cycle-to-cycle variance compounds over repeated assimilation updates. Ledoit–Wolf's low-variance estimator achieves the lowest EnKF RMSE on cosmological data despite having the worst standalone Frobenius loss.

The remainder of the paper is organised as follows. Section 2 reviews the EnKF, the Wishart distribution of sample covariances, and the main families of shrinkage estimators. Section 3 presents the direct precision shrinkage framework, the four targets we evaluate, and the proposed scaled identity target. Section 4 describes the dual-domain benchmark and reports results. Section 5 analyses the sequential accumulation effect and its implications for estimator design. Section 6 concludes.

2 Preliminaries

2.1 The Ensemble Kalman Filter

The Ensemble Kalman Filter [5] represents the background state with an ensemble of N_e state vectors $\{\mathbf{x}_i^b\}_{i=1}^{N_e}$, each $\mathbf{x}_i^b \in \mathbb{R}^n$, whose mean $\bar{\mathbf{x}}^b = N_e^{-1}\sum_i \mathbf{x}_i^b$ is the forecast estimate. Writing the deviation matrix $\Delta\mathbf{X}^b = \mathbf{X}^b - \bar{\mathbf{x}}^b\mathbf{1}^T \in \mathbb{R}^{n\times N_e}$, the sample background covariance is

$$\mathbf{S} \;=\; \mathbf{P}^b \;=\; \frac{1}{N_e-1}\,\Delta\mathbf{X}^b(\Delta\mathbf{X}^b)^T \;\in\; \mathbb{R}^{n\times n}. \tag{1}$$

The stochastic analysis solves $[\mathbf{H}\mathbf{P}^b\mathbf{H}^T + \mathbf{R}]\,\Delta\mathbf{X}^a = \mathbf{D}^s$ for the deviation of the analysis ensemble, with $\mathbf{D}^s$ the matrix of perturbed innovations. The *dual* formulation [18] uses the background *precision* directly,

$$\left[(\mathbf{P}^b)^{-1} + \mathbf{H}^T\mathbf{R}^{-1}\mathbf{H}\right]\Delta\mathbf{X}^a \;=\; \mathbf{H}^T\mathbf{R}^{-1}\mathbf{D}^s, \tag{2}$$

and is the entry point for this work: plugging a shrinkage-based precision estimator into (2) gives a drop-in replacement for the sample precision $\mathbf{S}^{-1}$ at each assimilation cycle.

2.2 The Wishart Distribution and Sample Precision

If the ensemble members are drawn from a Gaussian with true covariance $\boldsymbol{\Sigma}$, then $(N_e-1)\mathbf{S} \sim \mathcal{W}_n(N_e-1, \boldsymbol{\Sigma})$ is Wishart-distributed. Two properties matter for what follows. First, the eigenvalues of $\mathbf{S}$ are systematically spread relative to $\boldsymbol{\Sigma}$: the largest are overestimated, the smallest underestimated, and the bias grows with n/N_e. Inverting $\mathbf{S}$ turns the overestimated small eigenvalues into underestimated large eigenvalues of $\mathbf{S}^{-1}$—and vice versa—so the sample precision inherits the spreading with amplified noise on the small-N_e end. Second, when $N_e-1 > n+1$, $\mathbb{E}[\mathbf{S}^{-1}] = \frac{N_e-1}{N_e-n-2}\,\boldsymbol{\Sigma}^{-1}$, so $\mathbf{S}^{-1}$ is a biased estimator. The Hartlap

correction $\gamma = (N_e - n - 2)/(N_e - 1)$ [9] removes this bias in standalone use, but the Bodnar et al. [2] direct-precision shrinkage coefficients already account for it through the $(1 - n/N_e)$ term, so it should *not* be pre-applied when using those formulas.

2.3 Covariance Matrix Shrinkage

Covariance shrinkage blends $\mathbf{S}$ with a structured target $\mathbf{T}$,

$$\hat{\mathbf{P}} \;=\; (1-\lambda)\,\mathbf{S} \;+\; \lambda\,\mathbf{T}, \qquad \lambda \in [0,1], \tag{3}$$

and chooses λ to minimize $\mathbb{E}\|\hat{\mathbf{P}} - \boldsymbol{\Sigma}\|_F^2$. Ledoit and Wolf [14] derived a closed-form data-driven intensity for $\mathbf{T} = \mu\mathbf{I}$, $\mu = \mathrm{tr}(\mathbf{S})/n$, that is asymptotically optimal within this class and has become a standard reference. Chen et al. [3] refined the result with an oracle-approximating variant that is particularly effective in the small-N_e regime. Both estimators produce well-conditioned covariances but still require an explicit inversion to obtain a precision matrix, and that inversion amplifies errors along the smallest sample eigenvalues—precisely the directions where shrinkage was supposed to regularize.

NERCOME. NERCOME [13,12] avoids the target-choice problem by splitting the ensemble into disjoint subsets, diagonalizing the first subset's sample covariance, rotating the second subset into that eigenbasis, and keeping only the diagonal. Averaging over B random splits debiases the eigenvalue estimates nonparametrically. The price is $\mathcal{O}(B)$ eigendecompositions per estimate, which becomes prohibitive inside a sequential filter.

2.4 Precision Matrix Shrinkage

Rather than shrinking $\mathbf{S}$ and inverting, Bodnar et al. [2] proposed shrinking the precision matrix *directly*. The estimator combines the sample precision $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$ with a structured precision target $\boldsymbol{\Pi}_0$,

$$\boldsymbol{\Pi}_{\mathrm{LS}} \;=\; \alpha^*\,\hat{\boldsymbol{\Pi}}_S \;+\; \beta^*\,\boldsymbol{\Pi}_0, \tag{4}$$

with analytically derived coefficients

$$\alpha^* = 1 - \frac{n}{N_e} - \frac{N_e^{-1}\,\|\hat{\boldsymbol{\Pi}}_S\|_{\mathrm{tr}}^2\,\|\boldsymbol{\Pi}_0\|_F^2}{\|\hat{\boldsymbol{\Pi}}_S\|_F^2\,\|\boldsymbol{\Pi}_0\|_F^2 - \big(\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S\boldsymbol{\Pi}_0)\big)^2}, \tag{5}$$

$$\beta^* = \left(1 - \frac{n}{N_e} - \alpha^*\right)\frac{\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S\boldsymbol{\Pi}_0)}{\|\boldsymbol{\Pi}_0\|_F^2}, \tag{6}$$

that minimize the expected quadratic loss $\mathbb{E}\|\boldsymbol{\Pi}_{\mathrm{LS}} - \boldsymbol{\Sigma}^{-1}\|_F^2$ under the Wishart assumption. The denominator in (5) is non-negative by Cauchy–Schwarz and vanishes only when $\hat{\boldsymbol{\Pi}}_S \propto \boldsymbol{\Pi}_0$; when the target is a close match, the correction term drives α^* down and β^* up (more weight to the target), and vice versa. In practice the unclamped coefficients can leave $[0,1]$, in which case Bodnar et al. [2] clamp them before use.

Cosmological precision structure. For power-spectrum measurements of a Gaussian random field [8],

$$\mathrm{Cov}(\hat{C}_{\ell_i}, \hat{C}_{\ell_j}) \;=\; \frac{2}{(2\ell_i+1)f_{\mathrm{sky}}}\left(C_{\ell_i} + \frac{1}{n_g}\right)^2 \delta_{\ell_i \ell_j}, \tag{7}$$

is diagonal by mode independence. Pope and Szapudi [21], as adopted by Looijmans et al. [15], use the leading-order cosmic-variance part as a precision target via $T^{(2)}_{ii} = (2/N_{\ell_i})\, C^2_{\ell_i}$, with $N_{\ell_i} = (2\ell_i+1)f_{\mathrm{sky}}$, and $\boldsymbol{\Pi}^{(2)}_0 = (\mathbf{T}^{(2)})^{-1}$.

2.5 Localization and Modified Cholesky

In the data-assimilation community, the dominant remedy for rank-deficient ensemble covariances is *localization*: element-wise multiplication of $\mathbf{S}$ by a compactly supported correlation function [7,11] that zeros out long-range correlations. The *modified Cholesky decomposition* [18] is a banded variant that factors $(\mathbf{P}^b)^{-1} \approx \mathbf{L}^{-T}\mathbf{D}^{-1}\mathbf{L}^{-1}$ with $\mathbf{L}$ lower triangular and $\mathbf{D}$ diagonal, estimating each row of $\mathbf{L}$ by regressing a state variable on its neighbors within a bandwidth r. The cost is $\mathcal{O}(n\, r\, N_e)$ per cycle, linear in n for fixed r, which is the only scaling in our comparison that survives operational state dimensions. Localization and shrinkage are complementary: localization exploits spatial structure when it is known, shrinkage exploits similarity to a target when no such structure is available.

2.6 Critical Comparison of Existing Approaches

The estimators reviewed above address the same problem from complementary angles, but none is uniformly preferable, and their limitations jointly define the gap this paper addresses.

Shrinkage in covariance space is not shrinkage in precision space. Ledoit and Wolf [14] minimizes $\mathbb{E}\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_F^2$, not the precision loss that governs the EnKF update; the inverse of a covariance-optimal estimator amplifies errors in small eigenvalues, echoing the classical ridge-regression observation that regularizing a matrix and its inverse are different problems [10]. Cosmology has explored surrogate precision priors directly [20,23,22,4], but without porting them to sequential filtering.

Target-free methods pay for their generality. NERCOME removes the target-choice problem at the cost of $\mathcal{O}(B)$ eigendecompositions per cycle. If a reasonable target exists, a single-pass linear shrinkage is cheaper *and* lower-variance; if no target is available, NERCOME's cost becomes prohibitive inside the EnKF loop (see Section 4.4).

Diagonal targets assume the problem away. The identity, eigenvalue, and cosmological targets available in the direct-precision framework are all diagonal. This is appropriate for data whose true covariance is near-diagonal by construction (Gaussian random fields at independent Fourier modes) but not for dynamical systems with spatial coupling, where the true precision is *dense* and

off-diagonal. Shrinking toward a diagonal target then forces the Bodnar formulas to trade off two wrong ingredients, producing miscalibrated coefficients that we see directly in Section 4.4.

Standalone optimality is not sequential optimality. Every estimator above is designed for a single-shot loss. Inside an EnKF, the precision is recomputed at every cycle, and errors accumulate over hundreds of updates. A low-bias, high-variance estimator that dominates in a single-shot comparison can easily be outperformed over 100 cycles by a higher-bias, lower-variance alternative; this effect is the central empirical finding of this paper.

3 Proposed Method

3.1 Direct Precision Shrinkage for the EnKF

The Bodnar et al. [2] framework was developed for standalone precision matrix estimation and applied to cosmological data by Looijmans et al. [15]. Neither work embedded the estimator inside a sequential filter. We do so here: at each assimilation cycle of the dual EnKF (2), we replace the raw sample precision $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$ with the shrinkage estimator

$$\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0, \tag{8}$$

where $\boldsymbol{\Pi}_0$ is a structured target and the coefficients α^*, β^* are computed from Eqs. (5)–(6) at every cycle using the current ensemble.

The weight α^* on the sample precision is bounded above by $1 - n/N_e$, reflecting the known positive bias of the Wishart inverse. The remaining weight β^* on the target depends on how well $\boldsymbol{\Pi}_0$ aligns with $\hat{\boldsymbol{\Pi}}_S$: when $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)$ is large relative to $\|\boldsymbol{\Pi}_0\|_F^2$, more weight goes to the target. Both coefficients are clamped to $[0, 1]$ with $\alpha^* + \beta^* \leq 1$ as a finite-sample safeguard, since the asymptotic guarantees of Bodnar et al. do not hold at the ensemble sizes typical of data assimilation ($N_e = 20$–100).

Embedding this estimator in the EnKF introduces a consideration absent from standalone estimation: the precision matrix is recomputed at every assimilation cycle, and errors in the analysis state propagate through the forecast model into subsequent cycles. An estimator that is accurate on average but variable from cycle to cycle can accumulate larger errors over a long assimilation window than a slightly biased but stable one. This sequential accumulation effect is not captured by one-shot metrics like Frobenius loss and must be evaluated through filtering experiments (Section 4).

3.2 Shrinkage Targets

The four target variants below differ only in their choice of $\boldsymbol{\Pi}_0$. The first three follow Looijmans et al. [15]; the fourth is a domain-agnostic adaptation we propose.

Algorithm 1 Direct Precision Shrinkage (Identity Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute sample covariance: $\mathbf{S} = \frac{1}{N_e - 1} \mathbf{DX}\,\mathbf{DX}^T$
2: Compute sample precision: $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$
3: Set target: $\boldsymbol{\Pi}_0 = \mathbf{I}_n$
4: Compute α^*, β^* via Eqs. (5)–(6)
5: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0$

Algorithm 2 Direct Precision Shrinkage (Eigenvalue Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute $\mathbf{S}$, $\hat{\boldsymbol{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: Compute eigenvalues $e_1 \leq \cdots \leq e_n$ of $\hat{\boldsymbol{\Pi}}_S$
3: Set target: $\boldsymbol{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$
4: Compute α^*, β^* via Eqs. (5)–(6)
5: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0^{(1)}$

Identity Target. The simplest choice: $\boldsymbol{\Pi}_0 = \mathbf{I}_n$, assuming independent unit-variance priors. This carries no domain-specific information, but the direct precision formulation still produces a valid precision matrix—unlike inverting a shrunk covariance, which can amplify estimation error. Algorithm 1 summarises the procedure; the eigenvalue, cosmological, and scaled identity variants (Algorithms 2, 3, and Section 3.2) follow the same template with different choices of $\boldsymbol{\Pi}_0$.

Eigenvalue Target. A diagonal matrix of the sorted eigenvalues of $\hat{\boldsymbol{\Pi}}_S$: $\boldsymbol{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$ with $e_1 \leq \cdots \leq e_n$. This retains the spectral information of the sample precision while zeroing out off-diagonal correlations, and requires no external domain knowledge. It corresponds to the first precision target ($\boldsymbol{\Pi}_0^{(1)}$) in Looijmans et al. [15].

Cosmological Target. The analytical target from Looijmans et al., derived from the Gaussian power spectrum covariance (7). Each diagonal entry of the target covariance is $T_{ii}^{(2)} = (2/N_{\ell_i}) C_{\ell_i}^2$, encoding cosmic variance at multipole ℓ_i [8], and the target precision is its inverse: $\boldsymbol{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$. When the assumed power spectrum model is close to the truth, this provides a stronger shrinkage direction than the generic alternatives.

Scaled Identity Target (Novel Adaptation). The cosmological target requires power spectrum data C_ℓ and mode counts N_ℓ that are unavailable outside cosmology. We propose a domain-agnostic alternative that uses empirical per-variable variances from the sample covariance:

$$\boldsymbol{\Pi}_0 = \mathrm{diag}\left(\frac{1}{2\sigma_1^2}, \ldots, \frac{1}{2\sigma_n^2} \right), \tag{9}$$

Algorithm 3 Direct Precision Shrinkage (Cosmological Target)

Require: Deviation matrix $\mathbf{DX}$, power spectrum C_ℓ, mode counts N_ℓ
1: Compute $\mathbf{S}$, $\hat{\boldsymbol{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: **for** each data element i mapped to multipole ℓ_i **do**
3: Compute diagonal covariance target: $T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2$
4: **end for**
5: Set target precision: $\boldsymbol{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$
6: Compute α^*, β^* via Eqs. (5)–(6)
7: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0^{(2)}$

Table 1. Computational complexity per assimilation cycle. n: state dimension, N_e: ensemble size, r: Cholesky bandwidth, B: number of NERCOME bootstrap draws.

Method	Cost per cycle
EnKF (standard)	$O(n^2 N_e + m^2 N_e)$
EnKF-Cholesky	$O(n \cdot r \cdot N_e)$
Identity Shrinkage	$O(n^2 N_e + n^3)$
Scaled Shrinkage	$O(n^2 N_e + n^3)$
Cosmological Shrinkage	$O(n^2 N_e + n^3)$
Eigenvalue Shrinkage	$O(n^2 N_e + 2n^3)$
NERCOME	$O(B \cdot (n^2 N_e + n^3))$
Ledoit-Wolf	$O(n^2 N_e)$

where σ_i^2 are the diagonal entries of $\mathbf{S}$. The factor of 2 mirrors the $2/N_\ell$ scaling of the cosmological target in Section 2.4. This variant tests how much of the direct precision shrinkage performance depends on domain-specific target knowledge versus the general framework itself.

3.3 Computational Complexity

All four direct precision shrinkage variants share a common computational core: forming the sample covariance $\mathbf{S}$ costs $O(n^2 N_e)$, and inverting it to obtain $\hat{\boldsymbol{\Pi}}_S$ costs $O(n^3)$. Computing α^* and β^* requires the Frobenius norm $\|\hat{\boldsymbol{\Pi}}_S\|_F^2$, the trace norm $\|\hat{\boldsymbol{\Pi}}_S\|_{\mathrm{tr}}$, and the product $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)$, each costing $O(n^2)$ for a dense target or $O(n)$ for a diagonal one. The final linear combination is $O(n^2)$. Since the $O(n^3)$ inversion dominates, all four variants have the same asymptotic cost per assimilation cycle.

They differ in target construction. The identity target requires no computation. The scaled identity reads n diagonal entries of $\mathbf{S}$, costing $O(n)$. The cosmological target similarly costs $O(n)$ given precomputed C_ℓ and N_ℓ values. The eigenvalue target requires a symmetric eigendecomposition of $\hat{\boldsymbol{\Pi}}_S$, costing $O(n^3)$, which doubles the effective cost of that variant.

Table 1 summarizes the per-cycle cost of each method tested in this work, including the baselines.

The modified Cholesky approach avoids the $O(n^3)$ inversion entirely by exploiting banded structure, making it the cheapest option on spatially extended systems where $r \ll n$. Ledoit-Wolf shrinks the covariance directly and does not require inversion ($O(n^2 N_e)$), though the subsequent EnKF solve still involves an $O(m^2 N_e)$ linear system. NERCOME is the most expensive: its B bootstrap resamples (typically $B = 200$) each require a covariance computation and eigendecomposition, resulting in a cost roughly B times that of the other shrinkage methods. On the AR(1) model ($n = 18$, $N_e = 30$), NERCOME takes ∼27 seconds per 100 cycles versus ∼0.2 seconds for the other shrinkage methods. On Lorenz 96 ($n = 40$, $N_e = 80$), the gap widens to ∼215 seconds versus ∼26 seconds.

All implementations are publicly available in TEDA[1].

4 Benchmark

4.1 Experimental Setup

We test on two systems with different precision matrix structures.

Lorenz 96. The Lorenz 96 model [16] has $n = 40$ state variables with forcing $F = 8$, producing chaotic dynamics with dense off-diagonal precision structure due to spatial coupling. We use $m = 32$ randomly placed observations with Gaussian error $\sigma_o = 0.01$, multiplicative inflation $\rho = 1.05$, and 100 assimilation cycles at observation frequency $\Delta t_{\text{obs}} = 0.1$. Ensemble sizes are $N_e \in \{50, 60, 80, 100\}$, all above the $N_e > n + 2 = 42$ threshold required for direct precision methods. Each configuration is repeated 30 times with different random seeds.

AR(1) cosmological model. To test on data with near-diagonal precision structure, we build an AR(1) model calibrated to BOSS DR12 Patchy mock catalog statistics from Looijmans et al. [15]:

$$\mathbf{x}_{t+1} = \phi\, \mathbf{x}_t + (1 - \phi)\, \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{10}$$

with $\boldsymbol{\mu}$ the mean power spectrum from the mocks, $\phi = 0.95$, and $\mathbf{Q} = (1 - \phi^2)\boldsymbol{\Sigma}_{\text{mock}}$ so the stationary distribution has covariance $\boldsymbol{\Sigma}_{\text{mock}}$. This preserves the near-diagonal precision structure within a dynamical system. We use $d = 18$ state variables (9 k-bins × 2 multipoles), $m = 14$ observations (≈ 78% coverage), $\sigma_o = 100$, inflation $\rho = 1.02$, and $N_e \in \{24, 30, 40, 50\}$.

We also replicate the standalone precision estimation experiment of Looijmans et al. using the same BOSS mock catalogs. For subsample sizes $n \in \{24, 30, 40, 50\}$, we draw $M = 100$ random subsets of n mocks, compute each estimator's precision matrix, and measure the relative Frobenius loss against the true precision (computed from all 2048 mocks with Hartlap correction [9]):

$$\mathcal{L}(\hat{\boldsymbol{\Pi}}, \boldsymbol{\Pi}_{\text{true}}) = \frac{\|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}_{\text{true}}\|_F}{\|\boldsymbol{\Pi}_{\text{true}}\|_F}. \tag{11}$$

[1] TEDA source code: https://github.com/enino84/TEDA.git

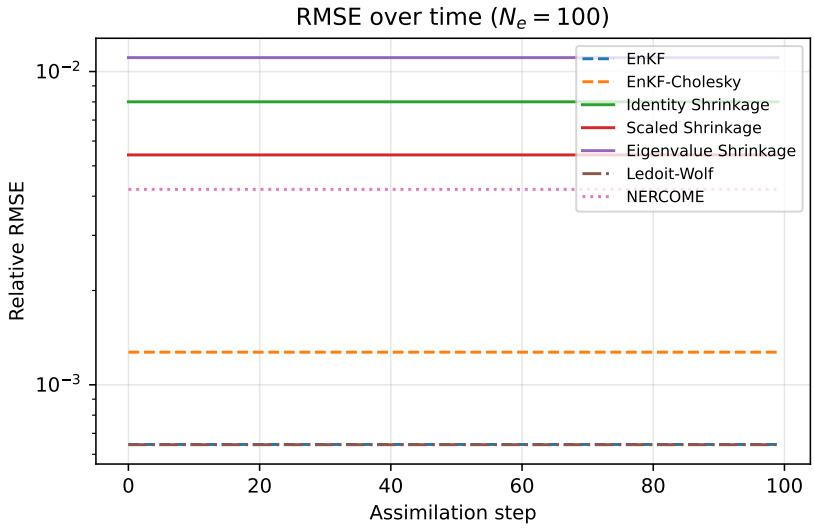

Fig. 1. Evolution of relative RMSE over 100 assimilation cycles for different analysis methods with $N_e = 100$. Shrinkage-based methods (solid lines) and Ledoit-Wolf (dash-dot) are compared against the standard EnKF and EnKF-Cholesky (dashed).

Seven methods are compared on Lorenz 96: standard EnKF, EnKF-Cholesky (bandwidth $r = 2$), three direct precision shrinkage variants (identity, scaled identity, eigenvalue targets), NERCOME, and Ledoit-Wolf. The cosmological target is excluded from Lorenz 96 since this model does not produce power spectrum data; it is evaluated on the cosmological experiments alongside all other methods.

For filtering experiments, we report the relative RMSE of the analysis state:

$$\mathrm{RMSE}(t) = \frac{\|\bar{\mathbf{x}}^a(t) - \mathbf{x}^{\mathrm{true}}(t)\|_2}{\|\mathbf{x}^{\mathrm{true}}(t)\|_2}, \tag{12}$$

time-averaged over the last 50 cycles to discard the initial transient.

4.2 Lorenz 96 Results

Figure 1 shows RMSE over 100 assimilation cycles for six methods at $N_e = 100$.

Table 2 summarizes performance across ensemble sizes. Ledoit-Wolf achieves the lowest RMSE at three of four ensemble sizes, with near-zero variance across runs. The direct precision shrinkage methods show erratic non-monotonic behavior—RMSE sometimes jumps at intermediate ensemble sizes before dropping again—and high standard deviations reflecting seed-dependent instability.

This instability traces to the Bodnar coefficients. While α^* stays near $1-n/N_e$ across all ensemble sizes and assimilation steps, β^* is severely miscalibrated:

Table 2. Time-averaged relative RMSE ($\times 10^{-2}$) for different methods and ensemble sizes on Lorenz 96. Best result per column in **bold**. †Scaled Shrinkage is a novel domain-agnostic adaptation proposed in this work.

Method	$N_e = 50$	$N_e = 60$	$N_e = 80$	$N_e = 100$
EnKF (standard)	3.06 ± 11.45	1.15 ± 4.33	0.08 ± 0.02	$\mathbf{0.06 \pm 0.01}$
EnKF-Cholesky	0.13 ± 0.01	0.14 ± 0.01	0.13 ± 0.01	0.13 ± 0.01
Identity Shrinkage	0.84 ± 0.57	1.13 ± 1.19	0.85 ± 0.46	0.80 ± 0.65
Scaled Shrinkage†	0.51 ± 0.28	0.73 ± 0.62	0.61 ± 0.33	0.54 ± 0.36
Eigenvalue Shrinkage	1.72 ± 0.83	1.81 ± 1.58	1.43 ± 1.01	1.11 ± 0.89
NERCOME	0.23 ± 0.15	0.28 ± 0.15	0.34 ± 0.17	0.42 ± 0.18
Ledoit-Wolf	$\mathbf{0.09 \pm 0.01}$	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.07 \pm 0.01}$	0.06 ± 0.01

unclamped values reach 10^8–10^9 for the identity target and 10^3–10^5 for the scaled target, requiring clamping to $[0, 1]$ at every step. For the eigenvalue target, β^* is effectively zero ($\sim 10^{-4}$), so no shrinkage toward the target occurs. The identity target $\boldsymbol{\Pi}_0 = \mathbf{I}$ is a poor match for Lorenz 96, whose precision matrix has significant off-diagonal entries from spatial coupling. In the formula for β^* (6), the $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)/\|\boldsymbol{\Pi}_0\|_F^2$ factor becomes very large when the sample precision has large diagonal entries relative to the target norm, driving β^* to extreme values. After clamping, the resulting precision matrices have condition numbers of 10^7–10^{11}, making the analysis updates numerically unstable.

Figure 2 plots the time-averaged RMSE against N_e. The standard EnKF and Ledoit-Wolf improve monotonically with larger ensembles. Modified Cholesky gives stable though not optimal performance across all sizes.

NERCOME shows a counterintuitive trend: RMSE *increases* with ensemble size (0.23 at $N_e = 50$ to 0.42 at $N_e = 100$). At small N_e, the sample covariance is noisy and bootstrap resampling provides genuine regularization; at larger N_e, the sample covariance is already reasonable, and bootstrap variance from 200 draws per cycle compounds over 100 sequential updates.

Figure 3 compares the precision matrices produced by each method at a representative assimilation step.

4.3 Cosmological Results

Standalone Precision Estimation. Table 3 shows the relative Frobenius loss for standalone precision matrix estimation on BOSS DR12 mock data. The results reverse compared to Lorenz 96: the direct precision shrinkage methods now outperform Ledoit-Wolf by a wide margin. At $n = 50$, the identity, eigenvalue, and cosmological targets all achieve losses of ≈ 0.41 versus 0.93 for Ledoit-Wolf. This confirms the findings of Looijmans et al. that direct precision shrinkage works well when the true precision matrix is near-diagonal.

Figure 4 visualizes these trends. Ledoit-Wolf's loss stays nearly flat at ≈ 0.93–0.96 regardless of subsample size: it shrinks the *covariance* toward a diagonal target and then inverts, a suboptimal route when the precision matrix itself is

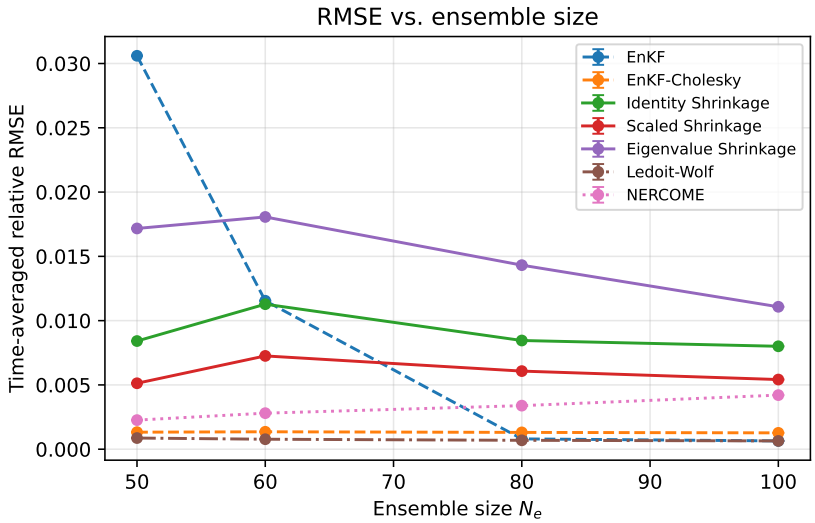

Fig. 2. Time-averaged relative RMSE (over last 50 cycles) as a function of ensemble size N_e on Lorenz 96. Error bars show ± 1 standard deviation over 30 runs.

near-diagonal. NERCOME does best at small n thanks to its low variance, and the scaled identity target achieves the lowest mean loss at $n = 30$. Both are overtaken by the Bodnar methods once the sample size is large enough for the sample precision to become reliable.

EnKF Filtering. Table 4 reports the time-averaged RMSE for the AR(1) cosmological model. Ledoit-Wolf achieves the lowest RMSE at all four ensemble sizes, despite having the worst standalone precision estimates. Unlike on Lorenz 96, the direct precision shrinkage methods now improve smoothly and monotonically with ensemble size, confirming that the erratic Lorenz 96 behavior was caused by target-truth mismatch, not a fundamental flaw.

Figure 5 plots RMSE against ensemble size. All methods converge as N_e grows. Pairwise Welch's t-tests ($n = 30$ runs) show that on Lorenz 96, Ledoit-Wolf's advantage is statistically significant ($p < 0.05$) at most ensemble sizes. On the cosmological data, most shrinkage methods do not differ significantly from Ledoit-Wolf ($p > 0.05$) at $N_e \geq 40$.

The scaled identity target occupies a middle ground. On Lorenz 96, it outperforms the other direct precision variants (Table 2), probably because scaling by empirical variances gives a better-conditioned target than the plain identity. On cosmological data, it achieves the lowest mean Frobenius loss at $n = 30$ (Table 3) but falls slightly behind the Bodnar methods at larger sample sizes. It serves its intended purpose: modest gains over the identity target without requiring external information, but no substitute for a domain-specific target when one is available.

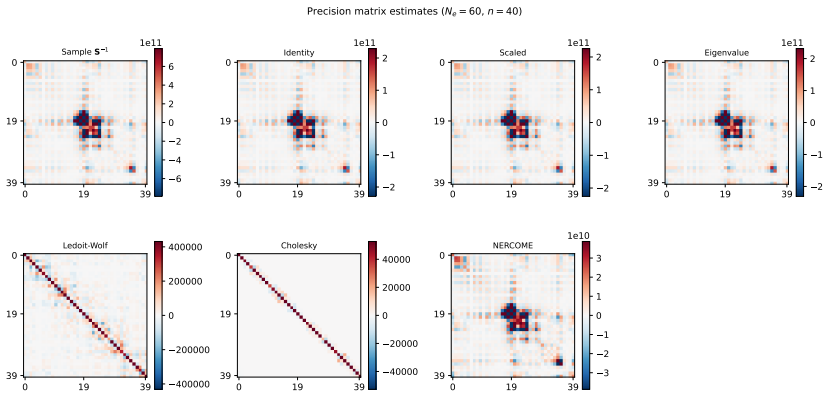

Fig. 3. Precision matrix estimates (40×40) from different methods with $N_e = 60$: sample precision $\mathbf{S}^{-1}$, identity shrinkage, scaled shrinkage, eigenvalue shrinkage, Ledoit-Wolf, and modified Cholesky. The direct precision shrinkage estimators produce matrices with much larger magnitude than Ledoit-Wolf and Cholesky, reflecting the ill-conditioning discussed above.

Table 3. Relative Frobenius loss of precision matrix estimates on BOSS DR12 mock data ($d = 18$). Lower is better. Best result per column in **bold**.

Method	$n = 24$	$n = 30$	$n = 40$	$n = 50$
Sample (Hartlap)	1.15 ± 1.01	0.74 ± 0.44	0.47 ± 0.15	0.43 ± 0.16
Identity Shrinkage	1.18 ± 1.13	0.68 ± 0.43	$\mathbf{0.45 \pm 0.12}$	$\mathbf{0.41 \pm 0.14}$
Scaled Shrinkage†	0.90 ± 0.96	$\mathbf{0.60 \pm 0.36}$	0.51 ± 0.11	0.46 ± 0.13
Eigenvalue Shrinkage	1.20 ± 1.13	0.70 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
Cosmological Shrinkage	1.19 ± 1.13	0.69 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
NERCOME	$\mathbf{0.82 \pm 0.06}$	0.73 ± 0.08	0.63 ± 0.08	0.53 ± 0.08
Ledoit-Wolf	0.96 ± 0.01	0.95 ± 0.01	0.94 ± 0.01	0.93 ± 0.01

4.4 Computational Cost

All shrinkage methods except NERCOME are within 15% of the EnKF baseline cost when $n \leq 40$. On the cosmological domain ($d = 18$, $N_e = 30$), NERCOME takes 26.7 s per 100 cycles, roughly 190× slower than the EnKF baseline (0.14 s) and 100× slower than the direct precision shrinkage variants (0.20–0.28 s). On Lorenz 96 ($n = 40$, $N_e = 80$), ensemble propagation dominates and the gap narrows: all methods except NERCOME take 26–28 s, with NERCOME at 215 s. Extrapolating to operational dimensions, the $\mathcal{O}(n^3)$ inversion that the direct precision methods share with Ledoit-Wolf becomes prohibitive at $n \gtrsim 10^4$; beyond that scale, only the modified Cholesky $\mathcal{O}(n\, r\, N_e)$ variant remains tractable without additional approximation.

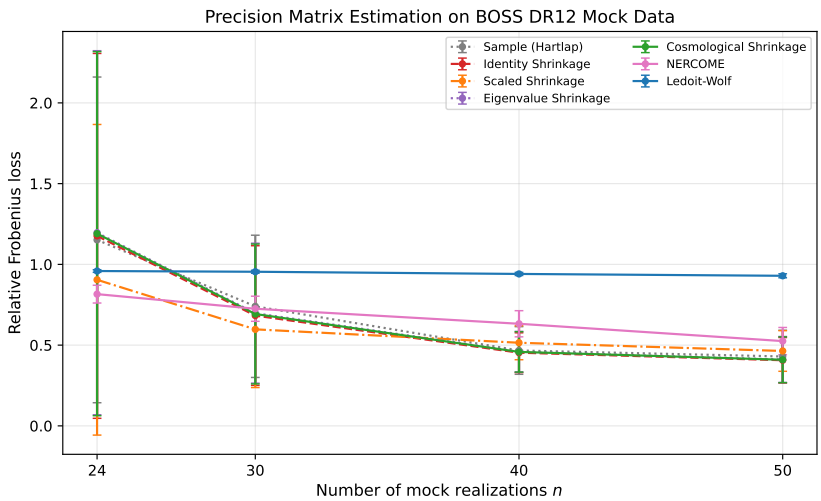

Fig. 4. Relative Frobenius loss of precision matrix estimates vs. subsample size on BOSS DR12 mock data ($d = 18$). Direct precision shrinkage methods (solid lines) achieve the lowest loss at $n \geq 40$, while Ledoit-Wolf (blue) remains flat near 0.93. Error bars show ± 1 standard deviation over 100 random draws.

5 Discussion

The benchmark in Section 4 produced two headline findings that run against the standard intuition for shrinkage estimation: (i) the best standalone precision estimator (direct precision shrinkage with a matched target) is not the best sequential estimator (Ledoit–Wolf), and (ii) direct precision shrinkage with a diagonal target fails spectacularly on a system with dense off-diagonal precision (Lorenz 96) even though the framework is formally correct. This section develops the explanation.

5.1 Target-Truth Alignment

The reversal between Lorenz 96 and cosmological data is driven by whether the target $\boldsymbol{\Pi}_0$ captures the dominant structure of the true precision. On BOSS-like data the true precision is near-diagonal by construction, and any of the diagonal targets we tested (identity, eigenvalue, cosmological, scaled identity) are a reasonable match; the Bodnar coefficients stay in $[0, 1]$, very little clamping occurs, and the method performs as the theory predicts. On Lorenz 96 the true precision is *dense* because nearest-neighbor coupling produces pervasive off-diagonal structure, yet all four targets are diagonal by construction. The Cauchy–Schwarz denominator in Eq. (5) becomes small relative to the numerator, β^* explodes to 10^8–10^9 for the identity target and 10^3–10^5 for the scaled variant, and clamping to $[0, 1]$ fires in over 80% of cycles. The resulting precision matrix is a clamped

Table 4. Time-averaged relative RMSE ($\times 10^{-2}$) for the AR(1) cosmological model. Best result per column in **bold**. †Scaled Shrinkage is a novel adaptation proposed in this work.

Method	$N_e = 24$	$N_e = 30$	$N_e = 40$	$N_e = 50$
EnKF (standard)	2.75 ± 0.46	2.32 ± 0.39	2.17 ± 0.20	2.11 ± 0.34
EnKF-Cholesky	1.99 ± 0.29	1.81 ± 0.26	1.87 ± 0.25	1.88 ± 0.31
Identity Shrinkage	1.95 ± 0.25	1.80 ± 0.26	1.88 ± 0.21	1.86 ± 0.30
Scaled Shrinkage†	1.99 ± 0.29	1.81 ± 0.26	1.89 ± 0.23	1.87 ± 0.31
Eigenvalue Shrinkage	2.35 ± 0.33	2.05 ± 0.30	2.00 ± 0.20	1.97 ± 0.32
Cosmological Shrinkage	2.21 ± 0.30	2.06 ± 0.31	2.04 ± 0.21	2.02 ± 0.33
NERCOME	1.88 ± 0.20	1.82 ± 0.19	1.91 ± 0.29	1.85 ± 0.27
Ledoit-Wolf	$\mathbf{1.88 \pm 0.26}$	$\mathbf{1.76 \pm 0.26}$	$\mathbf{1.86 \pm 0.21}$	$\mathbf{1.84 \pm 0.30}$

blend of two mismatched ingredients, with condition numbers of 10^7–10^{11}, and the EnKF update amplifies the error.

The eigenvalue target is a diagnostic counter-example: because it is derived from the same sample as $\hat{\boldsymbol{\Pi}}_S$, the Cauchy–Schwarz term $\|\hat{\boldsymbol{\Pi}}_S\|_F^2 \, \|\boldsymbol{\Pi}_0^{(1)}\|_F^2 - (\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0^{(1)}))^2$ is tiny and drives $\beta^* \approx 10^{-4}$. The estimator collapses back to $\alpha^* \hat{\boldsymbol{\Pi}}_S$—a rescaled sample precision with no effective shrinkage—so the eigenvalue target's poor Lorenz 96 performance is a symptom of self-reference, not of the framework.

5.2 Variance, not Bias, Dominates in Sequential Filtering

The more surprising observation is that Ledoit–Wolf wins the cosmological EnKF benchmark (Table 4) despite losing the standalone Frobenius-loss benchmark by a factor of two (Table 3). This is not an artefact: the standalone experiment gives each estimator one sample; the filtering experiment gives it one per cycle for 100 cycles, and the analysis error at step t feeds the forecast error at step $t+1$ through the model dynamics.

A simple accumulation argument makes this quantitative. If an estimator has per-step bias b and per-step variance s^2, the total error squared over K cycles scales roughly as $K\,b^2 + K\,s^2$ for approximately independent cycles. Ledoit–Wolf's standard deviation across ensemble draws is ± 0.01, an order of magnitude below the direct precision methods' ± 0.13–0.14. A $100\times$ reduction in s^2 easily outweighs a $2\times$ penalty in b^2 once K is in the hundreds. The bias–variance trade-off does not reverse between settings; the sequential setting merely re-weights it by the number of cycles.

This observation points to a gap in the shrinkage-estimation literature. Ledoit and Wolf [14], Chen et al. [3], and Bodnar et al. [2] all optimise single-shot loss. Review articles of the EnKF literature describe the sequential structure but do not connect it to estimator choice; the consequences are typically absorbed into inflation tuning rather than estimator selection. Our results suggest that, for sequential use, the *stability* of the estimator (its variance across realisations) is

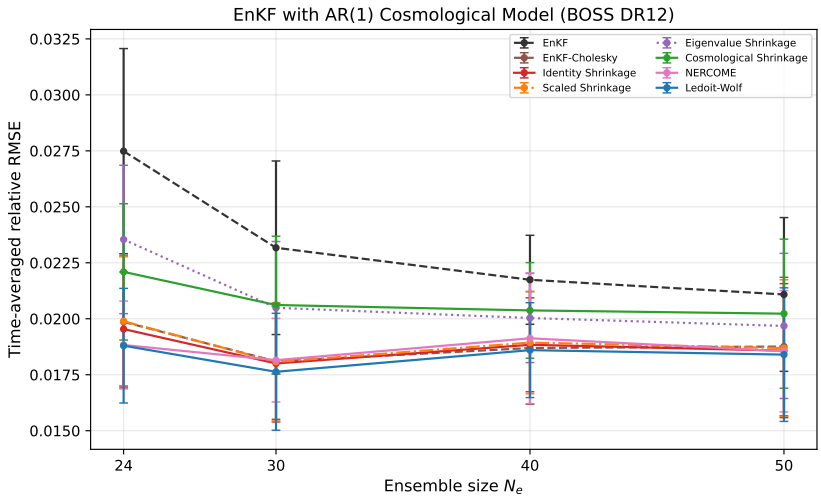

Fig. 5. Time-averaged relative RMSE vs. ensemble size for the AR(1) cosmological model. All methods show stable, monotonic behavior—unlike the Lorenz 96 case (Figure 2). Ledoit-Wolf achieves the lowest RMSE throughout. Error bars show ±1 standard deviation over 30 runs.

at least as important as its accuracy, and that this criterion is not visible in a standalone comparison.

5.3 Practical Recommendations

Combining the accuracy and cost results, a small decision table emerges. When $N_e \leq n$ (rank-deficient), only localization and the modified Cholesky variant apply; direct precision shrinkage is undefined. When $N_e \approx n$ (ill-conditioned) with near-diagonal precision and a known domain-specific target, direct precision shrinkage with that target gives the best standalone estimate; for sequential use, Ledoit–Wolf is the safer choice because of its low cycle-to-cycle variance. When the precision is dense, Ledoit–Wolf (or EnKF-Cholesky at intermediate n) is preferred; diagonal direct-precision targets fail. When $N_e \gg n$ (well-sampled), the sample precision is already accurate, and all shrinkage methods converge toward it. In all cases, NERCOME is ruled out for sequential use by its $\mathcal{O}(B)$ eigendecomposition cost.

The scaled identity target we propose occupies the niche where the precision is near-diagonal but no domain-specific target is available. It gives modest gains over the plain identity without requiring external information, and matches or beats Ledoit–Wolf on standalone estimation at small n. Its ad-hoc factor of 2, borrowed by analogy from the cosmological $2/N_\ell$ term, is admittedly heuristic and could likely be improved through cross-validation or a Bayesian argument.

5.4 Limitations and Future Work

Three limitations matter for interpreting the results. First, all experiments use a linear observation operator; nonlinear operators would introduce additional approximation error that can interact with the precision estimation error. Second, both test systems are low-dimensional ($n = 40$ and $d = 18$) compared to operational data assimilation ($n \sim 10^6$–10^8); the cost extrapolation in Section 4.4 implies that the $\mathcal{O}(n^3)$ inversion becomes prohibitive at $n \gtrsim 10^4$ unless the direct-precision framework is combined with localization or a low-rank approximation. Third, the sequential-accumulation effect is empirical; a rigorous Lyapunov-style analysis of how estimator variance propagates across filtering cycles would put our finding on firmer theoretical ground.

These limitations suggest three concrete extensions. A hybrid that first localises the sample covariance and then applies direct precision shrinkage could carry the framework to $n \sim 10^3$–10^4, though the localized matrix is no longer Wishart and the Bodnar derivation would need revisiting. An adaptive target that switches between diagonal and covariance-space methods based on a cheap off-diagonal-energy diagnostic would handle both Lorenz-like and cosmology-like regimes within a single estimator. And a condition-number-capped variant of the Bodnar estimator, truncating tail eigenvalues before coefficient computation, would directly address the failure mode observed on Lorenz 96.

6 Conclusions

We embedded the Bodnar et al. [2] direct precision shrinkage framework inside the dual Ensemble Kalman Filter, recomputing the shrinkage coefficients at each assimilation cycle, and benchmarked six estimators on two systems with opposite precision structure. Three findings emerge.

First, target-truth alignment determines direct precision shrinkage performance. With near-diagonal cosmological data, all four diagonal targets we tested yield well-behaved coefficients and the methods achieve the lowest standalone Frobenius loss. With dense-precision Lorenz 96 data, the same methods produce coefficients that require clamping in more than 80% of cycles, β^* values that drift to 10^8–10^9, and condition numbers of 10^7–10^{11}. The failure is a direct consequence of matching a diagonal target to a dense truth, not a flaw in the Bodnar et al. [2] framework.

Second, good standalone estimation does not imply good filtering. Ledoit–Wolf produces the worst standalone precision estimates on cosmological data (Frobenius loss ≈ 0.93–0.96 versus 0.41–0.46 for the direct precision methods) yet achieves the lowest EnKF RMSE at every tested ensemble size. Ledoit–Wolf's cycle-to-cycle variance is an order of magnitude below the direct precision methods', and over 100 sequential assimilation cycles that stability outweighs the bias penalty. For sequential data assimilation, variance reduction is at least as important as bias reduction.

Third, the scaled identity target we propose, $\boldsymbol{\Pi}_0 = \mathrm{diag}(1/(2\sigma_i^2))$, fills a middle ground between the plain identity (no scale information) and the cosmologi-

cal target (requires domain knowledge). It gives modest improvements over the plain identity target on both test systems without requiring any domain-specific information, and serves as a reasonable default for near-diagonal precision when no domain-specific target is available.

Practical guidance for choosing an EnKF precision estimator follows from these findings: in the absence of a domain-specific target that matches the true precision structure, Ledoit–Wolf is the safest default by virtue of its low cycle-to-cycle variance; direct precision shrinkage is preferable when a matched target exists *and* standalone estimation quality is the objective; NERCOME is ruled out for sequential use by its $\mathcal{O}(B)$ cost; and the modified Cholesky decomposition remains the only tractable option at operational state dimensions. Future work includes a hybrid that combines localization with direct precision shrinkage, an adaptive target selection mechanism that diagnoses the off-diagonal structure of the sample precision, and a theoretical treatment of the sequential accumulation effect via a Lyapunov analysis of the filter error. All implementations are publicly available in the TEDA toolbox[2].

Acknowledgements. This work was supported by the Department of Computer Science and Statistics at Universidad del Norte.

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